

BUSA 8031 Business Analytics Project
Personal Report Task 1

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Introduction

This report reflects my Job Simulation through Forage by completing the Tata Data Visualization tasks, which focus on applying data analytics to reach business decisions. In order to communicate the main issues brought up by the CEO and CMO, the simulation required analyzing the provided online retail data and creating interesting and significant visualizations. During the simulation, I was required to clean and prepare the dataset. This included completing the finding of significant criteria, the cleansing of the dataset, and the creation of visualisations that showcase insights from both the operational and marketing areas. Therefore, this process has the capacity to communicate not only technical information by cleansing and managing the dataset, but also the ability to interpret business requirements and translate them into practical insights.

The goal of this report is to discuss the business challenge that was encountered during the simulation, as well as the steps, tools, and differences from Tata's recommendations. Furthermore, this report will explore significant takeaways and identify areas for further growth to more effectively prepare for a career in the business analytics industry.

Business Problems

The primary issue of this simulation transforms business questions into data analysis, as the CEO and CMO have different perspectives on the company. The CEO is more concerned with operational aspects and overall business success, such as revenue and expansion potential, whereas the CMO is primarily concerned with customer insights, purchasing patterns, and high-value customers. Additionally, since the simulation expected the participant to clean up the dataset, the initial dataset contained some data that, if not handled carefully, could lead to an error, such as negative numbers and wrong unit prices. This purification process taught me the value of data accuracy for future analysis.

Eventually, to successfully communicate the analysis, the usage of visualisation becomes the simulation's challenge. As the analysis was intended for senior management, it was important to present the data straightforwardly and understandably. This required not only technical knowledge of Tableau but also an understanding of how to present information to support business decision-making.

Tasks, Steps, Tools

To solve the business problems in the simulation, a structured method was used, beginning with data preparation and progressing to analysis and visualisation. To ensure accuracy, the dataset was cleaned in Microsoft Excel first. Records with negative quantities (Figure 1) and unit prices less than or equal to zero (Figure 2) were removed since they may affect the results. This process provided that the insights gained were reliable.

InvoiceNo	InvoiceDate	Quantity	UnitPrice	CustomerID	Country
536355	5/1/2010 08:45	6	52.5	13185	United Kingdom
536355	5/1/2010 08:45	6	20.34	13185	United Kingdom
536355	5/1/2010 08:45	22	22	13185	United Kingdom
536355	5/1/2010 08:45	20.34	20.34	13185	United Kingdom
536355	5/1/2010 08:45	20.34	20.34	13185	United Kingdom
536355	5/1/2010 08:45	20.34	20.34	13185	United Kingdom
536355	5/1/2010 08:45	20.34	20.34	13185	United Kingdom
536355	5/1/2010 08:45	20.34	20.34	13185	United Kingdom
536355	5/1/2010 08:45	20.34	20.34	13185	United Kingdom
536355	5/1/2010 08:45	20.34	20.34	13185	United Kingdom

Figure 1: Quantity column filtering

InvoiceNo	InvoiceDate	Quantity	UnitPrice	CustomerID	Country
536355	5/1/2010 08:45	6	52.5	13185	United Kingdom
536355	5/1/2010 08:45	6	20.34	13185	United Kingdom
536355	5/1/2010 08:45	22	22	13185	United Kingdom
536355	5/1/2010 08:45	20.34	20.34	13185	United Kingdom
536355	5/1/2010 08:45	20.34	20.34	13185	United Kingdom
536355	5/1/2010 08:45	20.34	20.34	13185	United Kingdom
536355	5/1/2010 08:45	20.34	20.34	13185	United Kingdom
536355	5/1/2010 08:45	20.34	20.34	13185	United Kingdom
536355	5/1/2010 08:45	20.34	20.34	13185	United Kingdom
536355	5/1/2010 08:45	20.34	20.34	13185	United Kingdom

Figure 2: UnitPrice column filtering

After cleaning, a revenue column was created by multiplying quantity by unit price, resulting in a consistent measure of business performance.

Tableau was then used to create visualizations that answered the CEO's and CMO's questions. A line chart was built to show the monthly income trend for 2011 (Figure 3), which helped in identifying seasonal patterns.

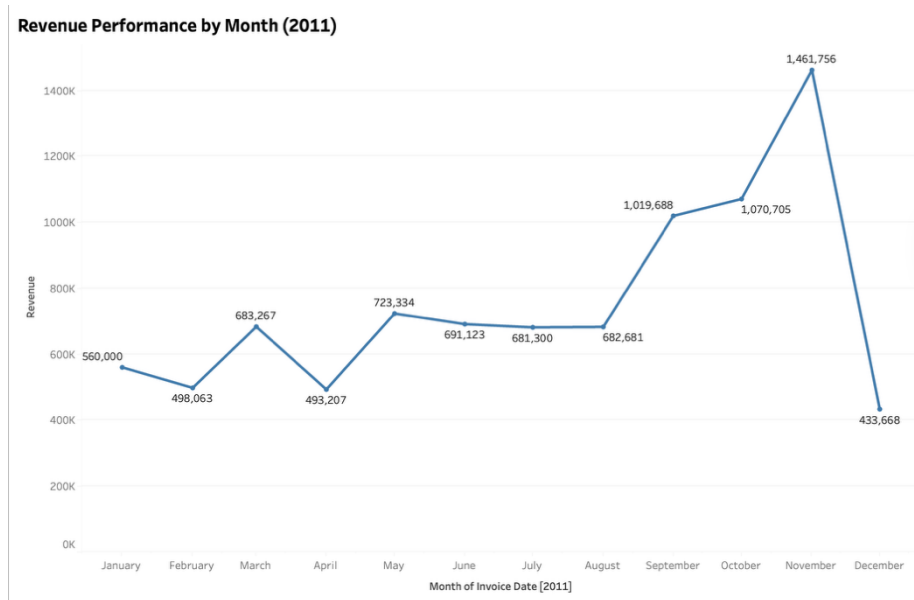


Figure 3: Monthly Revenue Performance (2011)

The figures indicate that income has been reasonably consistent over the first eight months, averaging roughly \$600,000 - \$700,000 each month. However, revenue begins to rise in September, up around 42% - 45% from the previous month, and continues to rise until it reaches a peak of over \$1.5 million in November. There can be no clear answer because the December data looks to be incomplete. Overall, the results show an important seasonality near the end of the year.

As a response to the CMO's request, a bar chart was created to identify the top ten countries by revenue, excluding the United Kingdom (Figure 4). The quantity sold was also included to provide more perspective.

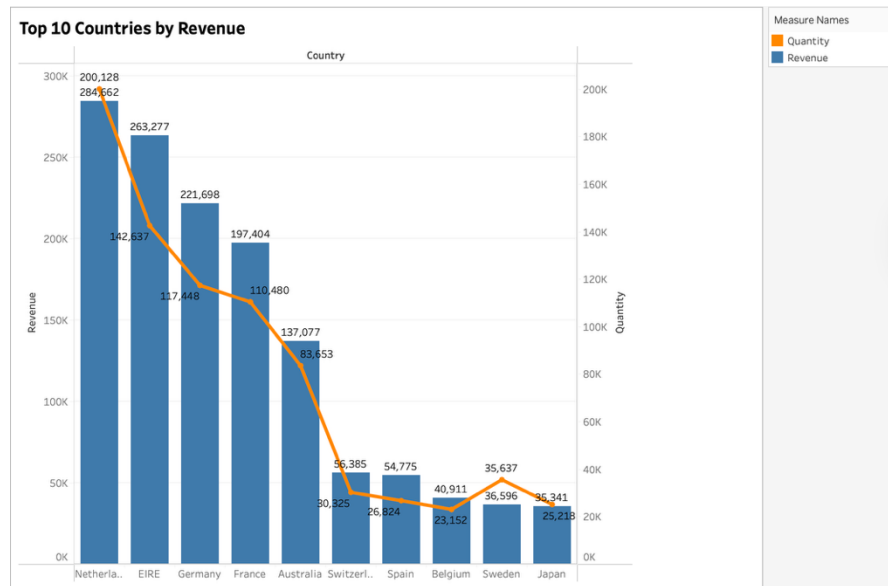


Figure 4: Top 10 countries by revenue generating (excluding the United Kingdom)

The findings reveal that a small number of nations account for a significant share of total revenue. In general, larger revenue correlates with a higher amount sold, showing strong demand. Sweden, on the other hand, is an outlier in that a bigger quantity does not necessarily equate to a higher revenue. This may indicate that lower-priced products were purchased in higher quantities.

To further analyze customer performance, a bar chart was created to identify the top 10 customers by revenue (Figure 5).

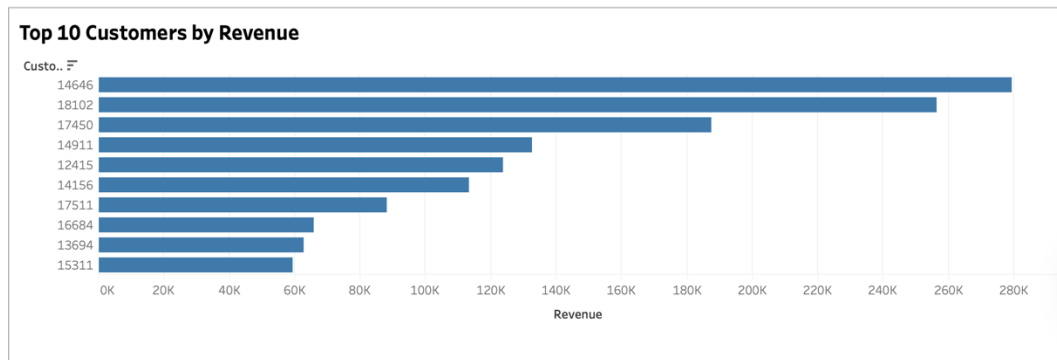


Figure 5: Top 10 customers by revenue generating

According to the visualization, the data show that a small group of customers represents a significant portion of total revenue. For example, the top customer generates significantly more revenue than the others, followed by a sharp decline among the remaining consumers. This trend suggests that the business is slightly dependent on a small number of high-value customers. While these customers are important for income generation, they can also cause a problem if the company becomes too reliant on them. From a marketing perspective, this highlights the necessity of retaining these high-value customers through customized strategies, as well as growing the customer base to reduce dependence and support long-term growth.

In order to explore potential expansion opportunities, product demand in various countries was evaluated using quantity sold, excluding the United Kingdom (Figure 6). This method highlights areas with the most demand.

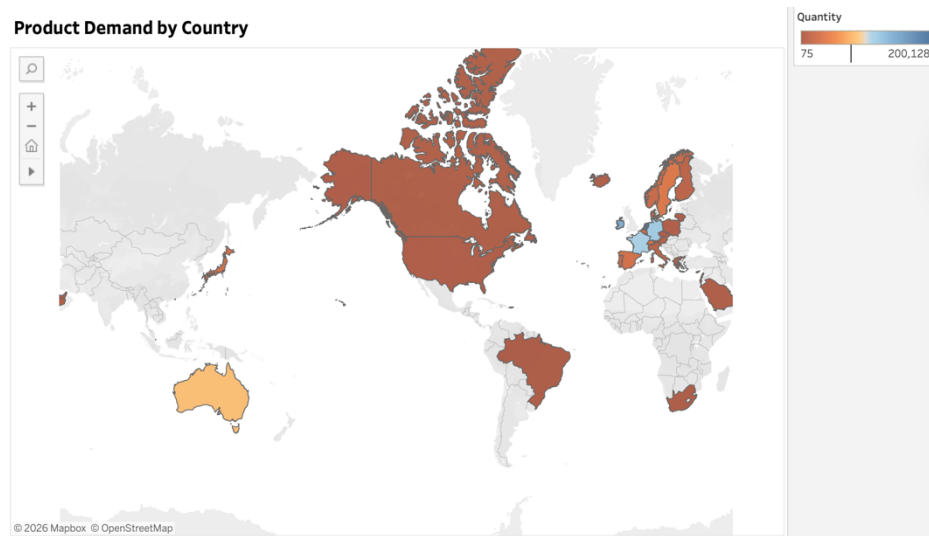


Figure 6: Intensity based on quantity demanded in each country

The visualization compares product demand across countries based on quantity sold, allowing for the identification of places with stronger demand. The color scale is key in understanding the data, with darker blue areas representing higher demand and lighter orange tones indicating lower demand. The visual shows France and Germany highlighted in darker blue, indicating that they have the highest levels of demand among the countries analyzed. Countries portrayed in lighter tones, such as Australia, have significantly lower demand. This shows that France and Germany are important markets for the company and may present significant growth potential. Focusing on these regions through better distribution and focused strategies may assist in maximizing future growth.

Comparison with Tata

Overall, my approach was similar to Tata's, especially with the use of visualisations to answer business questions. Similar to Tata, I used line charts, bar charts, and maps to display revenue patterns, compare performance across geographies and customers, and evaluate product demand. These visualisations helped the CEO and CMO make informed decisions. However, while Tata's method focused on presenting clear and structured findings, my analysis focused on a deeper examination of the data. For example, I found Sweden to be a situation where more quantity did not result in higher revenue, revealing differences in pricing or purchasing behaviour. In addition, I discussed the possible risk of relying on a small number of customers of high value, which goes beyond basic performance analysis and takes into account business sustainability.

This comparison shows that, while my work is consistent with Tata's organised analytical methodology, I was able to develop a more critical and business-oriented perspective by detecting patterns, exceptions, and potential risks in the data.

Key Takeaways

1. Importance of Data Handling and Insight Generation

The importance of handling data correctly before performing any analysis is a key lesson to be learned from this simulation. Negative amounts and incorrect unit pricing were among the dataset's problems, which, if ignored, could produce inaccurate findings. This experience made it clear that the first step in ensuring accuracy is data cleaning. It also highlighted how important it is to go beyond basic data and focus on getting insightful information that might help with business decisions.

2. Selection of the appropriate visualization

Another important takeaway is the relevance of using the right visualisation to successfully communicate ideas. Different types of data require different visual representations, such as line charts for trends or bar charts for comparisons. This simulation focused on the importance of selecting the appropriate visualisation to ensure that information is clear, easy to interpret, and valuable for decision-making.

3. Acknowledgement of stakeholder needs

The simulation additionally highlighted the need to know the needs of many stakeholders. The CEO was primarily concerned with overall performance and growth potential, but the CMO was more engaged in consumer behaviour and market performance. This involved adjusting the analysis and visualisations to reflect varied perspectives, underscoring the importance of effective data analysis for business outcomes.

Future Development

1. Strengths developed

Through this simulation, I have become better at handling, analysing, and using visualization to turn data into insightful conclusions. I gained additional confidence in my ability to clean datasets, find relevant metrics, and analyse outcomes from a business perspective. As an example, I was able to spot trends like revenue seasonality and exceptions like Sweden's distinct purchasing habits. My ability to transform unprocessed data into useful insights that assist in business decision-making has improved as a result of this experience.

2. Areas for improvement

Despite these improvements, I recognise that there is still potential for improvement. Developing my analytical abilities to go beyond simple interpretation and use advanced techniques is one important area. Additionally, I would like to get better at communicating insights, especially when it involves explaining complicated material to many stakeholders in an organized and understandable manner.

3. Further development

Moving further, I intend to continue developing my technical and communication skills in order to satisfy the demands of the employment market. This includes developing my skills with technologies like Tableau and others, and broadening my knowledge of advanced analytical methodologies. At the same time, I intend to improve my capacity to express ideas clearly and effectively in both written and verbal communication. Overall, this simulation has helped me understand more fully the abilities required for a business analytics career and has influenced my future growth objectives.